Covering the plane by rotations of a periodic stripe: the covering radius, its elementary theory, and an open lower bound

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Abstract

For $\varepsilon > 0$ let $E(\varepsilon)$ be the union of the closed vertical stripes of half-width ε centred at the integers, and for a real angle θ let R_θ denote the rotation of the plane about the origin through θ. The Pyjama problem asks whether finitely many rotations of $E(\varepsilon)$ can cover the plane. Manners answered this in the affirmative for every $\varepsilon > 0$, but his argument does not control the number $N(\varepsilon)$ of rotations needed; Kravitz and Leng gave the first explicit quantitative bound, of the very weak shape $\exp\exp\exp(\varepsilon^{-C})$, while a volume count gives only $N(\varepsilon) \gtrsim \varepsilon^{-1/2}$.

We isolate one scalar attached to a finite set Θ of angles, its *covering radius* $r(\Theta)$, which turns coverage into the single inequality $r(\Theta) \le \varepsilon$. We develop its elementary theory in ten results, each with a complete stepwise proof: the covering criterion, the fact that the defining function evaluated at a grid point is a rigorous lower bound for $r(\Theta)$ at any mesh, a Lipschitz bound on a box, a period-lattice reduction that upgrades a box bound to a global one whenever two independent exact periods exist, the universal bound $r(\Theta) \le 1/2$, monotonicity in Θ, the exact value $r(\Theta) = 1/2$ for a single direction and for any pair of independent directions, and the conditional consequence that $r(\Theta) \ge 1/(2|\Theta|)$ would improve the volume lower bound from $\varepsilon^{-1/2}$ to $1/(2\varepsilon)$. We state that inequality as a conjecture: it is open for $|\Theta| \ge 3$, and we say precisely which step is missing.

Numerical evidence is deferred to Appendix A, where the algorithms, the parameter domain, the readings and the provenance of every statement are collected. The conjecture is there reported as supported by computation and not proved.

1 Introduction

1.1 The problem

Let

$$E(\varepsilon) \;=\; \{\, z \in \mathbb{R}^2 \,:\, \operatorname{dist}(z_1, \mathbb{Z}) \le \varepsilon \,\}, \tag{1}$$

the union of the closed vertical stripes of half-width ε centred on the integers, and for a real angle θ let R_θ be the rotation of $\mathbb{R}^2$ about the origin through θ. The Pyjama problem asks for which $\varepsilon > 0$ there is a *finite* set $\Theta = \{\theta_1, \dots, \theta_k\}$ with

$$\bigcup_{i=1}^{k} R_{\theta_i} E(\varepsilon) \;=\; \mathbb{R}^2. \tag{2}$$

One rotation never suffices, since a rotation of the stripe family leaves all stripes parallel to one another; the question is how many directions are needed. Its quantitative form asks for the smallest admissible k, written $N(\varepsilon)$, as a function of ε.

The problem is not a matter of area, because the stripes are unbounded and their images overlap in a complicated way: a point is uncovered exactly when, in every one of the k rotated frames, its coordinate lies at distance more than ε from an integer. Writing $u_\theta = (\cos\theta, \sin\theta)$, the quantity

$$f_\Theta(z) \;=\; \min_{1\le i\le k} \operatorname{dist}(z\cdot u_{\theta_i}, \mathbb{Z}) \tag{3}$$

therefore measures how far z is from being covered in the worst direction, and coverage (2) is the statement that f_Θ nowhere exceeds ε. Making that equivalence into a usable tool — including the exact constant it carries — is the content of Section 3.

1.2 What the literature settles

The problem was posed by Iosevich, Kolountzakis and Matolcsi [IKM06]. Besides the planar stripe question they prove a negative result in the lattice analogue: for a discrete lattice Λ and ε smaller than half the length of the shortest non-zero vector of Λ, no finite family of rotations produces a (Λ,ε)-good covering, and they leave the periodic stripe arrangement itself open.

Manners [Man15] settled the finiteness question: for every $\varepsilon > 0$, finitely many rotations of $E(\varepsilon)$ cover the plane. His method recasts the question as a dynamical statement on an appropriate solenoid; it establishes existence and does not track how the number of rotations depends on ε.

Malikiosis, Matolcsi and Ruzsa [MMR13] treat variants of the problem, including solutions that are doubly periodic with respect to a common period lattice and the behaviour of families of rotations in general position; our description of that work follows its published abstract, as we did not have access to the full text.

Kravitz and Leng [KL25] give the first explicit quantitative upper bound: $\exp\exp\exp(\varepsilon^{-C})$ rotations suffice, for an absolute constant C. On the other side the volume count gives only $N(\varepsilon) \gtrsim \varepsilon^{-1/2}$, because k rotations of the stripe produce k stripes of width 2ε per period cell and their total area cannot exceed the area of the cell. The gap between $\varepsilon^{-1/2}$ and $\exp\exp\exp(\varepsilon^{-C})$ is the quantitative content of the problem, and it is wide open.

1.3 What this paper does

We do not attempt to close that gap. We isolate the scalar $r(\Theta)$ of Definition 2.2, the supremum of f_Θ, and we prove ten elementary results about it. Three of them are tools: any grid point gives a rigorous lower bound at any mesh (Lemma 3.3), a box bound follows from a Lipschitz estimate (Lemma 3.4), and a box bound becomes global as soon as the period group of Θ has rank two (Theorem 3.5). The others are structural: the covering criterion (Theorem 3.2), the universal bound $r(\Theta) \le 1/2$ together with monotonicity in Θ (Theorem 3.6), the exact values for one direction and for two independent directions (Propositions 3.7, 3.8 and Corollary 3.9), and the conditional consequence for $N(\varepsilon)$ (Proposition 3.11).

The structural results are proved in full and are unconditional. The lower bound $r(\Theta) \ge 1/(2|\Theta|)$ is not: it holds with equality for $|\Theta| \le 2$ and is stated as Conjecture 3.10 for $|\Theta| \ge 3$, where it is open. Section 4 says exactly which step is missing.

1.4 The status of the evidence

Every assertion of Section 3 is a proved mathematical statement, with a natural-language proof written out step by step so that each step can be checked line by line. The conjecture of Section 3 is supported by computation and by nothing more. All computational material — the procedure, the parameter domain, the readings, and the provenance identifiers — is collected in Appendix A;

numerical evidence is deferred to Appendix A and is used nowhere in the proofs of Section 3. No automated proof checker was applied to any proof in this paper.

2 Definitions and notation

Definition 2.1 (Stripe and rotation)**.** For $\varepsilon > 0$, $E(\varepsilon)$ is given by (1). For $\theta \in \mathbb{R}$ we write $u_\theta = (\cos\theta, \sin\theta) \in \mathbb{R}^2$ and R_θ for the rotation of $\mathbb{R}^2$ about the origin through the angle θ.

Definition 2.2 (Covering function and covering radius)**.** For a finite set $\Theta = \{\theta_1, \dots, \theta_k\} \subset \mathbb{R}$ and $p \in \mathbb{R}^2$,

$$f_\Theta(p) \;=\; \min_{1\le i\le k} \operatorname{dist}(p \cdot u_{\theta_i}, \mathbb{Z}), \qquad r(\Theta) \;=\; \sup_{p\in\mathbb{R}^2} f_\Theta(p).$$

We call f_Θ the covering function of Θ and $r(\Theta)$ its covering radius.

Definition 2.3 (Period group)**.** For a finite Θ, $\Lambda(\Theta) = \{\, v \in \mathbb{R}^2 \,:\, v \cdot u_\theta \in \mathbb{Z} \text{ for every } \theta \in \Theta \,\}$.

Definition 2.4 (Grid on a box)**.** Let $B = [a_1, b_1] \times [a_2, b_2]$ and let $n, m \ge 1$. The grid of mesh (h_x, h_y) with $h_x = (b_1 - a_1)/n$ and $h_y = (b_2 - a_2)/m$ is $\mathcal{G} = \{(a_1 + ih_x,\, a_2 + jh_y) : 0 \le i \le n,\ 0 \le j \le m\}$, and $L_{\mathcal{G}} = \max_{q\in\mathcal{G}} f_\Theta(q)$.

Both f_Θ and $r(\Theta)$ depend only on the set of directions $\{u_\theta : \theta \in \Theta\}$, so Θ and $\Theta + \pi$ give the same covering radius; we shall not distinguish such sets, and we understand Θ as an unordered set throughout.

3 Main results

We begin with the one-dimensional fact from which every constant below is inherited.

Lemma 3.1 (Sawtooth)**.** *Let $g(t) = \operatorname{dist}(t, \mathbb{Z})$. Then*

(i) $0 \le g(t) \le 1/2$ for every $t \in \mathbb{R}$, with $g(t) = 1/2$ if and only if $t \in \mathbb{Z} + \frac{1}{2}$;

(ii) $|g(t) - g(s)| \le |t - s|$ for all $s, t \in \mathbb{R}$.

Proof. (1) *The bound.* Let $t \in \mathbb{R}$ and $n = \lfloor t \rfloor$, so $n \le t \le n + 1$ and the nearest integer to t is n or $n + 1$. Hence $g(t) = \min(t - n,\, n + 1 - t)$. Both arguments are non-negative and they sum to 1, so their minimum is at most $1/2$; and $g(t) \ge 0$ by definition.

(2) *Equality.* If $\min(a, b) = 1/2$ with $a, b \ge 0$ and $a + b = 1$, then $a = b = 1/2$; here that reads $t - n = n + 1 - t = 1/2$, i.e. $t = n + \frac{1}{2}$. Conversely, for $t = n + \frac{1}{2}$ the two neighbours n and $n + 1$ of t in $\mathbb{Z}$ are both at distance $1/2$ and no integer is closer, so $g(t) = 1/2$.

(3) *Lipschitz.* Fix s and let $m \in \mathbb{Z}$ be a nearest integer to s, so that $|s - m| = g(s)$. Then $g(t) \le |t - m| \le |t - s| + |s - m| = |t - s| + g(s)$. Interchanging s and t gives $|g(t) - g(s)| \le |t - s|$. □

Theorem 3.2 (Covering criterion)**.** *For every finite $\Theta = \{\theta_1, \dots, \theta_k\}$ and every $\varepsilon > 0$,*

$$\bigcup_{i=1}^{k} R_{\theta_i} E(\varepsilon) = \mathbb{R}^2 \qquad \Longleftrightarrow \qquad r(\Theta) \le \varepsilon.$$

Proof. (1) *One stripe.* Fix $z \in \mathbb{R}^2$ and θ. Since R_θ is a bijection, $z \in R_\theta E(\varepsilon)$ if and only if $R_{-\theta} z \in E(\varepsilon)$, which by (1) means $\operatorname{dist}\big((R_{-\theta} z)_1, \mathbb{Z}\big) \le \varepsilon$. Now $R_{-\theta} z = (z_1 \cos\theta + z_2 \sin\theta,\ -z_1 \sin\theta + z_2 \cos\theta)$, so $(R_{-\theta} z)_1 = z \cdot u_\theta$ and

$$z \in R_\theta E(\varepsilon) \iff \operatorname{dist}(z \cdot u_\theta, \mathbb{Z}) \le \varepsilon.$$

(2) *The union.* A point lies in the union of the k sets if and only if it lies in one of them, so by (1) applied to each index i,

$$z \in \bigcup_{i=1}^{k} R_{\theta_i} E(\varepsilon) \iff \min_{1 \le i \le k} \operatorname{dist}(z \cdot u_{\theta_i}, \mathbb{Z}) \le \varepsilon \iff f_\Theta(z) \le \varepsilon.$$

(3) *Passing to the supremum.* Coverage is the statement that $f_\Theta(z) \le \varepsilon$ for every z, i.e. that ε is an upper bound for the set of values $f_\Theta(\mathbb{R}^2)$. The least upper bound of that set is $r(\Theta)$, so the statement is equivalent to $r(\Theta) \le \varepsilon$: if $r(\Theta) \le \varepsilon$ then $f_\Theta(z) \le r(\Theta) \le \varepsilon$ for every z; conversely, if $f_\Theta \le \varepsilon$ pointwise then ε is an upper bound and therefore dominates the least one. □

Lemma 3.3 (Grid points are rigorous lower bounds)**.** *Let Θ be finite, let $\mathcal{G}$ be a non-empty finite subset of $\mathbb{R}^2$ and $L_\mathcal{G} = \max_{q \in \mathcal{G}} f_\Theta(q)$. Then for every $q \in \mathcal{G}$,*

$$f_\Theta(q) \ \le\ L_\mathcal{G} \ \le\ r(\Theta).$$

In particular a maximum over a grid is a lower bound for $r(\Theta)$ at every mesh, however coarse, and refining a grid can only increase the bound.

Proof. (1) *The maximum is attained.* A non-empty finite set of real numbers has a maximum, so $L_\mathcal{G} = f_\Theta(q_0)$ for some $q_0 \in \mathcal{G}$; and $f_\Theta(q) \le L_\mathcal{G}$ for every $q \in \mathcal{G}$ by definition of a maximum.

(2) *The maximum is dominated by the supremum.* By Definition 2.2, $r(\Theta) = \sup_{p \in \mathbb{R}^2} f_\Theta(p)$ is an upper bound for all values of f_Θ, in particular $L_\mathcal{G} = f_\Theta(q_0) \le r(\Theta)$. Combining (1) and (2) gives the displayed chain.

(3) *Monotonicity under refinement.* If $\mathcal{G} \subseteq \mathcal{G}'$ then the maximum over $\mathcal{G}'$ is taken over a larger finite set of candidates, hence $L_\mathcal{G} \le L_{\mathcal{G}'}$. Thus the bound is valid at any mesh, and its *quality* depends on the mesh, which is what Lemma 3.4 quantifies. □

Lemma 3.4 (Lipschitz bound on a box, and its globalization)**.** *Let Θ be finite, let $B = [a_1, b_1] \times [a_2, b_2]$ and let $\mathcal{G}$ be the grid of Definition 2.4 of mesh (h_x, h_y). Then*

$$\sup_{p \in B} f_\Theta(p) \ \le\ L_\mathcal{G} + \tfrac{1}{2}\sqrt{h_x^2 + h_y^2}. \tag{4}$$

If moreover the closed box B is a fundamental domain for a period lattice of f_Θ of rank two, then

$$r(\Theta) \ \le\ L_\mathcal{G} + \tfrac{1}{2}\sqrt{h_x^2 + h_y^2}. \tag{5}$$

Proof. (1) *Each summand is 1-Lipschitz.* Fix a unit vector u and put $g_u(p) = \operatorname{dist}(p \cdot u, \mathbb{Z})$. For $p, q \in \mathbb{R}^2$ the Cauchy–Schwarz inequality gives $|p \cdot u - q \cdot u| = |(p-q) \cdot u| \le |p - q|$, and Lemma 3.1(ii) gives the 1-Lipschitz property of $\operatorname{dist}(\cdot, \mathbb{Z})$ on $\mathbb{R}$. Composing the two, $|g_u(p) - g_u(q)| \le |p \cdot u - q \cdot u| \le |p - q|$.

(2) *The minimum is 1-Lipschitz.* Write $g_i = g_{u_{\theta_i}}$. By (1), $g_i(p) \le g_i(q) + |p - q|$ for every i. Choose i_0 with $g_{i_0}(q) = f_\Theta(q)$; then $f_\Theta(p) \le g_{i_0}(p) \le g_{i_0}(q) + |p - q| = f_\Theta(q) + |p - q|$, and exchanging p and q yields $|f_\Theta(p) - f_\Theta(q)| \le |p - q|$.

(3) *Nearest grid point.* Let $p \in B$. Choose $i \in \{0, \dots, n\}$ and $j \in \{0, \dots, m\}$ with $|p_1 - (a_1 + ih_x)| \le h_x/2$ and $|p_2 - (a_2 + jh_y)| \le h_y/2$; such indices exist because $a_1 \le p_1 \le b_1$ and $a_2 \le p_2 \le b_2$. Put $q = (a_1 + ih_x,\, a_2 + jh_y) \in \mathcal{G}$. Then $|p - q|^2 \le h_x^2/4 + h_y^2/4$, that is $|p - q| \le \frac{1}{2}\sqrt{h_x^2 + h_y^2}$. By (2),

$$f_\Theta(p) \;\le\; f_\Theta(q) + \tfrac{1}{2}\sqrt{h_x^2 + h_y^2} \;\le\; L_{\mathcal{G}} + \tfrac{1}{2}\sqrt{h_x^2 + h_y^2}.$$

Taking the supremum over $p \in B$ gives (4).

(4) *Globalization.* Suppose B is a fundamental domain for a lattice Λ of exact periods of f_Θ, of rank two: every $z \in \mathbb{R}^2$ is $z = q + v$ with $q \in B$ and $v \in \Lambda$, and $f_\Theta(z) = f_\Theta(q)$. Then $r(\Theta) = \sup_{\mathbb{R}^2} f_\Theta = \sup_B f_\Theta$, so (4) gives (5). □

Theorem 3.5 (Period-lattice reduction)**.** *Let Θ be finite and let $\Lambda(\Theta)$ be as in Definition 2.3.*

(a) Every $v \in \Lambda(\Theta)$ is an exact period of f_Θ: $f_\Theta(p + v) = f_\Theta(p)$ for all p.

(b) If $\Lambda(\Theta)$ contains two linearly independent vectors v_1, v_2, then with $P = \{s_1 v_1 + s_2 v_2 : 0 \le s_1, s_2 < 1\}$ and $\overline{P}$ its closure,

$$r(\Theta) = \sup_{q \in \overline{P}} f_\Theta(q).$$

(c) Let $e_j = v_j/|v_j|$, let $h_1, h_2 > 0$, and let $\mathcal{G}$ be the set of points $jh_1 e_1 + lh_2 e_2$ $(j, l \in \mathbb{Z})$ at distance at most $\frac{1}{2}(h_1 + h_2)$ from $\overline{P}$. Then

$$\sup_{q \in \overline{P}} f_\Theta(q) \;\le\; L_{\mathcal{G}} + \tfrac{1}{2}(h_1 + h_2).$$

Proof. (1) *(a).* Let $v \in \Lambda(\Theta)$, $p \in \mathbb{R}^2$ and $1 \le i \le k$. Then $(p + v) \cdot u_{\theta_i} = p \cdot u_{\theta_i} + v \cdot u_{\theta_i}$ with $v \cdot u_{\theta_i} \in \mathbb{Z}$, and translating a real number by an integer does not change its distance to $\mathbb{Z}$: $\operatorname{dist}(p \cdot u_{\theta_i} + \text{integer}, \mathbb{Z}) = \operatorname{dist}(p \cdot u_{\theta_i}, \mathbb{Z})$. Taking the minimum over i gives $f_\Theta(p + v) = f_\Theta(p)$.

(2) *(b).* Let $p \in \mathbb{R}^2$. By linear independence, $p = a_1 v_1 + a_2 v_2$ for unique $a_1, a_2 \in \mathbb{R}$. Let $n_j = \lfloor a_j \rfloor$ and $v = n_1 v_1 + n_2 v_2$, an element of $\Lambda(\Theta)$ because it is an integer combination of v_1, v_2. Then $q = p - v = \{a_1\} v_1 + \{a_2\} v_2$ lies in P, where $\{\cdot\} \in [0, 1)$ denotes the fractional part, and by (a) $f_\Theta(p) = f_\Theta(q)$. Hence every value of f_Θ on $\mathbb{R}^2$ is attained on P; continuity of f_Θ and compactness of $\overline{P}$ give $r(\Theta) = \sup_{\mathbb{R}^2} f_\Theta = \sup_{\overline{P}} f_\Theta$.

(3) *(c).* Let $q \in \overline{P}$ and write $q = a_1 v_1 + a_2 v_2$ with $0 \le a_j \le |v_j|$. Choose integers j, l with $|a_1 - jh_1| \le h_1/2$ and $|a_2 - lh_2| \le h_2/2$, and put $g = jh_1 e_1 + lh_2 e_2 \in \mathcal{G}$; indeed $|q - g| \le \frac{1}{2}\sqrt{h_1^2 + h_2^2} \le \frac{1}{2}(h_1 + h_2)$, so g lies in the set defining $\mathcal{G}$. By Lemma 3.4(2) and that estimate,

$$f_\Theta(q) \;\le\; f_\Theta(g) + \tfrac{1}{2}(h_1 + h_2) \;\le\; L_{\mathcal{G}} + \tfrac{1}{2}(h_1 + h_2),$$

and taking the supremum over $q \in \overline{P}$ proves the claim. □

Theorem 3.6 (Universal bound and monotonicity)**.** *Let Θ be finite. Then*

(i) $r(\Theta) \le 1/2$;

(ii) if $\Theta \subseteq \Theta'$ then $r(\Theta) \le r(\Theta')$.

Proof. (1) *Pointwise bound.* Let $p \in \mathbb{R}^2$ and fix any index i_0. Since a minimum is at most any of its entries, $f_\Theta(p) \le \operatorname{dist}(p \cdot u_{\theta_{i_0}}, \mathbb{Z}) \le 1/2$ by Lemma 3.1(i).

(2) *Supremum.* The bound in (1) is uniform in p, so $r(\Theta) = \sup_{p \in \mathbb{R}^2} f_\Theta(p) \le 1/2$.

(3) *Monotonicity.* If $\Theta \subseteq \Theta'$, the minimum defining $f_{\Theta'}$ is taken over a superset of the indices defining f_Θ; hence $f_{\Theta'}(p) \le f_\Theta(p)$ for every p, and taking suprema of both sides over $\mathbb{R}^2$ preserves the inequality. □

Proposition 3.7 (A single direction)**.** *For every $\theta \in \mathbb{R}$, $r(\{\theta\}) = 1/2$.*

Proof. (1) *Upper bound.* $r(\{\theta\}) \le 1/2$ by Theorem 3.6(i).

(2) *Lower bound.* Take $p = \frac{1}{2}u_\theta$. Then $p \cdot u_\theta = \frac{1}{2}u_\theta \cdot u_\theta = \frac{1}{2}$, so by Lemma 3.1(i) $f_{\{\theta\}}(p) = \operatorname{dist}(\frac{1}{2}, \mathbb{Z}) = \frac{1}{2}$. Hence $r(\{\theta\}) \ge f_{\{\theta\}}(p) = 1/2$.

(3) *Conclusion.* The two inequalities give $r(\{\theta\}) = 1/2$. □

Proposition 3.8 (A pair of orthogonal directions)**.** *For $\Theta = \{0, \pi/2\}$ one has $r(\Theta) = 1/2$.*

Proof. (1) *Upper bound.* $r(\Theta) \le 1/2$ by Theorem 3.6(i).

(2) *Lower bound.* Here $u_0 = (1, 0)$ and $u_{\pi/2} = (0, 1)$. Take $p = (\frac{1}{2}, \frac{1}{2})$; then $p \cdot u_0 = p \cdot u_{\pi/2} = \frac{1}{2}$, so $f_\Theta(p) = \min(\frac{1}{2}, \frac{1}{2}) = \frac{1}{2}$ and $r(\Theta) \ge f_\Theta(p) = 1/2$.

(3) *Conclusion.* Together with (1), $r(\Theta) = 1/2$. □

Corollary 3.9 (Any two independent directions)**.** *Let u, v be linearly independent unit vectors and let Θ be the set of their arguments. Then $r(\Theta) = 1/2$.*

Proof. (1) *Solving for a point.* The linear map $T(p) = (p \cdot u,\, p \cdot v)$ has kernel $\{p : p \perp u \text{ and } p \perp v\}$, which is $\{0\}$ because u, v are independent in $\mathbb{R}^2$; hence T is an isomorphism and there is p with $T(p) = (\frac{1}{2}, \frac{1}{2})$.

(2) *Lower bound.* For that p, $f_\Theta(p) = \min(\operatorname{dist}(\frac{1}{2}, \mathbb{Z}), \operatorname{dist}(\frac{1}{2}, \mathbb{Z})) = \frac{1}{2}$, so $r(\Theta) \ge 1/2$.

(3) *Conclusion.* With Theorem 3.6(i), $r(\Theta) = 1/2$. □

So for $|\Theta| \le 2$ the covering radius equals $1/2$ exactly, and by Theorem 3.2 a set of at most two directions covers only at levels $\varepsilon \ge 1/2$. The following is the natural extrapolation.

Conjecture 3.10 (Lower bound in terms of the number of directions)**.** *For every finite Θ with $|\Theta| = k$,*

$$r(\Theta) \;\ge\; \frac{1}{2k}.$$

Proposition 3.7 and Corollary 3.9 verify Conjecture 3.10 with equality for $k = 1$ and $k = 2$. For $k \ge 3$ it is open; see Section 4. Its interest for the original problem lies in the following conditional statement.

Proposition 3.11 (Conditional lower bound for the number of rotations)**.** *Assume Conjecture 3.10. If Θ is finite and $\bigcup_{\theta\in\Theta} R_\theta E(\varepsilon) = \mathbb{R}^2$, then*

$$|\Theta| \;\ge\; \frac{1}{2\varepsilon}.$$

Proof. (1) *Coverage is an inequality.* By Theorem 3.2, coverage at level ε is equivalent to $r(\Theta) \le \varepsilon$.

(2) *Lower bound for the radius in terms of cardinality.* Let $k = |\Theta|$. For $k \le 2$ we have $r(\Theta) = 1/2 \ge 1/(2k)$ by Proposition 3.7 and Corollary 3.9; for $k \ge 3$ the same inequality is precisely Conjecture 3.10. In both cases $r(\Theta) \ge 1/(2k)$.

(3) *Combining.* Thus $1/(2k) \le r(\Theta) \le \varepsilon$, which gives $k \ge 1/(2\varepsilon)$. □

Remark 3.12 (What a computation can and cannot bracket). For a fixed Θ, Lemma 3.3 certifies a lower bound for $r(\Theta)$ from a single evaluation, at any mesh, while an upper bound is available only under a periodicity hypothesis, through Lemma 3.4 and Theorem 3.5(c). This asymmetry — a witness certifies from below, only a rank-two lattice of exact periods certifies from above — also governs the computational study of Conjecture 3.10: computation can refute the conjecture at a particular Θ by exhibiting one point, but it cannot confirm the conjecture, and what the readings of Appendix A confirm is the inequality $r(\Theta) \ge 1/(2k)$ for the sets examined, not the conjecture itself.

4 Discussion and open problems

4.1 What is established

By Theorem 3.2 the covering question for rotations of the periodic stripe is equivalent to a single scalar inequality: a finite Θ covers at level ε if and only if $r(\Theta) \le \varepsilon$. The covering radius is always at most $1/2$ (Theorem 3.6(i)), is monotone in the set of directions (Theorem 3.6(ii)), equals $1/2$ exactly for one direction and for any two independent directions (Propositions 3.7, 3.8, Corollary 3.9), and is computable in the two-sided sense of Lemma 3.3 and Theorem 3.5: certified from below by evaluation, and from above by evaluation together with a rank-two period lattice.

The general lesson is that the difficulty of the problem sits entirely in the lower bound: the upper bound $1/2$ is trivial, and every exact value we know equals $1/2$.

4.2 The conjecture and the missing step

Conjecture 3.10 asserts $r(\Theta) \ge 1/(2k)$ for $|\Theta| = k$. The cases $k \le 2$ are proved above, and in both of them an extremal point is produced in closed form: $p = \frac{1}{2}u_\theta$ for $k = 1$, and the solution of a 2×2 linear system for $k = 2$. For $k \ge 3$ neither device is available. What is missing is precisely the following:

- one needs, for an *arbitrary* k-element Θ, a point p with $\operatorname{dist}(p \cdot u_{\theta_i}, \mathbb{Z}) \ge 1/(2k)$ simultaneously for all i;
- a pointwise argument controls one direction at a time, since $\operatorname{dist}(t, \mathbb{Z}) \le 1/2$ for every t and a condition on the i-th coordinate says nothing about the remaining $k - 1$ coordinates;
- the natural simultaneous devices (Dirichlet-type approximation, or a product of k one-dimensional coverings) give constants strictly weaker than $1/(2k)$ and cannot be sharpened to it without an additional structural input on Θ;
- the extremal configurations — the sets Θ minimising $r(\Theta)$ for fixed k — are not identified for any $k \ge 3$.

Conjecture 3.10 should therefore be read as a statement about configurations, not about points.

4.3 Relation to the quantitative problem

The volume count gives $N(\varepsilon) \gtrsim \varepsilon^{-1/2}$, and the explicit bound of [KL25] is of size $\exp\exp\exp(\varepsilon^{-C})$. Proposition 3.11 shows that Conjecture 3.10 would yield $N(\varepsilon) \ge 1/(2\varepsilon)$: a polynomial improvement of the lower bound, still far from the triple-exponential upper bound. The gap is thus structural on both sides, and nothing known to us suggests that either bound is close to the truth.

4.4 Further questions

(Q1) Determine $r(\Theta)$ for the equispaced families $\Theta = \{i\pi/k : 0 \le i < k\}$. The exact value is known to us only for $k = 1, 2$; is it $1/(2k)$, i.e. is the equally spaced configuration extremal for every k?

(Q2) Is the infimum $\inf_{|\Theta|=k} r(\Theta)$ attained, and is it monotone in k? A failure of monotonicity would refute Conjecture 3.10 in a strong form.

(Q3) For which Θ does $\Lambda(\Theta)$ have rank two? Rank at most one is the generic situation for real angles, and it is exactly the situation in which the upper bound of Lemma 3.4 cannot be globalised; a classification would separate the computable cases from the incomputable ones.

(Q4) Extend Theorem 3.5(c) to a sub-lattice of $\Lambda(\Theta)$ up to a controlled error, which would give certified global upper bounds beyond rational multiples of π.

References

[IKM06] A. Iosevich, M. N. Kolountzakis and M. Matolcsi, *Covering the plane by rotations of a lattice arrangement of disks*, arXiv:math/0611800 (2006).

[Man15] F. Manners, *A solution to the pyjama problem*, Inventiones Mathematicae (2015), doi:10.1007/s00222-014-0571-7.

[MMR13] R.-D. Malikiosis, M. Matolcsi and I. Z. Ruzsa, *A note on the pyjama problem*, European Journal of Combinatorics **34** (2013), no. 7, 1071–1077.

[KL25] N. Kravitz and Leng, *Quantitative pyjama*, arXiv:2510.17744 (2025).

A Computational evidence

Everything in this appendix is computational or bookkeeping. Nothing here is used as a hypothesis in Section 3: the proofs there are self-contained.

A.1 The algorithm, stated as a mathematical procedure

Fix Θ with $|\Theta| = k$ and the threshold $\tau = 1/(2k)$. The procedure produces a two-sided bracket for $r(\Theta)$, as far as the available periodicity allows.

Step 1: lower bound. Choose the box $B = [-1, 1]^2$ and a grid with n subdivisions per axis, so $h = h_x = h_y = 2/n$. Evaluate f_Θ at every grid point and set $r_{\text{lower}} = L_{\mathcal{G}}$. By Lemma 3.3 this is a rigorous lower bound for $r(\Theta)$, valid for every Θ and every mesh.

Step 2: box-local upper bound. By Lemma 3.4,

$$\sup_{p \in B} f_\Theta(p) \;\le\; r_{\text{upper,box}} \;=\; L_{\mathcal{G}} + \tfrac{1}{2}\sqrt{2h^2} \;=\; L_{\mathcal{G}} + \frac{h}{\sqrt{2}}.$$

This is a statement about B only, and is recorded as such.

Step 3: global upper bound via the period lattice. Compute $\Lambda(\Theta) = \{v : v \cdot u_{\theta_i} \in \mathbb{Z}\ \forall i\}$ in exact rational arithmetic when the angles are rational multiples of π. If $\Lambda(\Theta)$ has rank two, replace B by a fundamental parallelogram P of $\Lambda(\Theta)$ and apply Theorem 3.5(c) with steps h_1, h_2; then $r_{\text{upper}} = L_{\mathcal{G}}(P) + \frac{1}{2}(h_1 + h_2)$ is a genuine global upper bound for $r(\Theta)$. If the rank is less than two, no such reduction is available and the only global upper bound is the trivial constant $1/2$ of Theorem 3.6(i).

Step 4: the verdict. The conjecture at level k is *supported* on Θ if $r_{\text{lower}} \geq \tau$, and *refuted* on Θ if $r_{\text{upper}} < \tau$ with a global upper bound from Step 3. A box-local upper bound below τ is not a refutation: by Lemma 3.3 a single point with $f_\Theta(p) < \tau$ would already be a counterexample, and such a point, if it exists inside B, is found by the evaluation of Step 1.

A.2 Parameter domain

The sweep covers five structural families of angle sets for each $k = 3, \ldots, 10$, giving $5 \times 8 = 40$ cells: equispaced $\{i\pi/k\}$, cyclic $\{2i\pi/k\}$, commensurable $\{2i\pi/(2k+1)\}$, random rational $\{a_i\pi/997\}$ with integer a_i, and random real angles. All cells are evaluated on $B = [-1, 1]^2$ with $n = 128$ subdivisions per axis, i.e. 129 points per axis and mesh $h = 1/64 = 0.015625$.

The exhaustive scan covers the N-th roots of unity for $N = 12$, $k = 3, \ldots, 8$ and for $N = 24$, $k = 3, 4, 5$, enumerating every k-element subset with no sampling:

$$\sum_{k=3}^{8} \binom{12}{k} + \sum_{k=3}^{5} \binom{24}{k} = 3718 + 55\,154 = 58\,872.$$

For each such Θ an explicit point p^* with $f_\Theta(p^*) \geq \tau$ is produced; by Lemma 3.3 this certifies $r(\Theta) \geq \tau$ rigorously for that Θ.

A.3 Readings

Table 1: Readings of the computational study. The two kinds of bound are of different strength: a certified lower bound is a rigorous statement about that particular Θ (Lemma 3.3), whereas a box-local upper bound is a statement about the box alone.

Quantity	Reading
Cells in the sweep ($k = 3, \ldots, 10$, five families)	40
Cells with certified lower bound $\geq 1/(2k)$	40 of 40
Cells with a global upper bound below $1/(2k)$	0
Certified refutations inside the searched box	0
Minimum certified lower bound over all cells	0.1367
Minimum box-local upper bound over all cells	0.1459
Maximum certified lower bound over all cells	0.5000
Angle sets in the exhaustive scan	58 872
Sets with an explicit witness $f(p^*) \geq 1/(2k)$	58 872
Hardest scanned set ($N = 12$, $k = 6$): lower bound / threshold	0.2113 / 0.0833
Numeric runs recorded	25
conclusive, supporting the conjecture	5
inconclusive	12
other verdicts, none read as a refutation	8

The sweep establishes no global statement: each of its 40 cells admits a certified lower bound at least $1/(2k)$, none admits a global upper bound below $1/(2k)$, and no counterexample was found inside the searched box. Its verdict is recorded as inconclusive, for lack of a global upper bound, and not because of a failure of the computation. The same reading applies to the exhaustive

scan: the certified inequalities hold for all 58 872 sets, and the number of box candidates with $r_{\mathrm{upper,box}} < 1/(2k)$ is 0.

Of the 25 recorded numeric runs, 5 are conclusive and support the conjecture, 12 are inconclusive, and the remaining 8 carry other verdicts. An inconclusive run is never read as evidence for the negation of the statement it failed to settle; in particular the runs along the Diophantine denseness route that exited with an error code are recorded as no conclusion.

A.4 Provenance

For auditability we list the internal identifiers of the material assembled for the project: the conjecture, its registered negation, the relations behind Section 1, and each computational run alluded to above. Identifiers appear only in this appendix.

Conjecture 3.10 `conj:000000`: the statement $r(\Theta) \geq 1/(2k)$ for $|\Theta| = k$, together with its registered negation and its refutation route.

Literature relations behind Section 1 `rel:000002`, `rel:000003`, `rel:000004`; `stmt:000020` (definition of $E(\varepsilon)$), `stmt:000021` (the problem as posed), `stmt:000022` (the affirmative answer), `stmt:000023` (the rationality lemma), `stmt:000025` (the original open problem).

Exact values at $k = 1, 2$ (Propositions 3.7, 3.8, Corollary 3.9) `exp:000021`, independently recomputed in `exp:000000` and `exp:000002`.

Period-lattice reduction (Theorem 3.5) `exp:000004`.

Supporting structural runs `exp:000006`, `exp:000012`, `exp:000014`; the counterexample recorded in `exp:000018` refutes an auxiliary intermediate claim of one attack route, not Conjecture 3.10.

The sweep of Table 1 `exp:000023`.

The exhaustive scan `exp:000024`.

Inconclusive runs `exp:000001`, `exp:000003`, `exp:000007`, `exp:000008`, `exp:000009`, `exp:000010`, `exp:000011`, `exp:000013`.

Literature `lit:000000` (Manners), `lit:000001` (Kravitz–Leng), `lit:000002` (Iosevich–Kolountzakis–Matolcsi), `lit:000003` (Malikiosis–Matolcsi–Ruzsa, used in Section 1 from its published abstract), `lit:000004` (the view-obstruction formulation, context only).

A.5 Evidence tier

The proofs of Section 3 are natural-language proofs, written out step by step so that every step can be checked line by line; they were reviewed by hand. No automated proof checker was applied to them. The computational readings of this appendix are cross-checks of individual instances: they support Conjecture 3.10 and they do not prove it, and the only statements of this paper that are proved are those of Section 3.